A COMPETITIVE ENVIRONMENT IN OUTER SPACE

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I. INTRODUCTION

In the 1920s, air transport was considered a fledgling industry, an industry in its infancy, which could only be kept "in the air" by direct or indirect State subsidies.[1] Today, air transport and the aeronautical activities related to it are mature industries, at least in most countries, and they are increasingly subject to ordinary business rules and laws, including competition and antitrust laws. Commercial activities in outer space – for convenience sake we will call them astronautical activities- are of much more recent vintage. They began, mostly in the form of telecommunication satellites, in the 1960s.[2] When the Outer Space Treaty[3] was being drafted during that same decade, the debate as to what "freedom of exploration and use by

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[1] The so-called Warsaw Convention of 1929, with its system of financial limits on air carriers' liability, is typically a recognition of the fact that the fledgling air transport industry could not yet support an ordinary system of civil liability without financial ceilings. Convention for the Unification of Certain Rules Relating to International Transportation by Air, Oct. 12, 1929, 49 Stat. 3014, 137 L.N.T.S. 11 [hereinafter Warsaw Convention]. *See also* P.P.C. HAANAPPEL, THE LAW AND POLICY OF AIR SPACE AND OUTER SPACE: A COMPARATIVE APPROACH 17 (Kluwer Law International, The Hague / London / New York, 2003).

[2] See Nicolas M. Matte, *Commercial and Cultural Utilization of Outer Space,* AEROSPACE LAW, FROM SCIENTIFIC EXPLORATION TO COMMERCIAL UTILIZATION (Carswell / Pedone, 1977).

[3] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies, Including the Moon and Other Celestial Bodies, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty].

all States"[4] of outer space meant, was still very much alive. Could "use" mean *commercial* use and could *private enterprise* participate in such commercial use of outer space? Eventually, over the initial objections of the Soviets, the answer adopted in Article VI of the Outer Space Treaty became "yes". Commercial uses of outer space and commercial uses of such space by private enterprise are permissible, as long as they are conducted under the authorisation and supervision of the State responsible for such private enterprise.[5]

Today, both commercialisation and privatisation of outer space activities are progressing rapidly, particularly in the areas of space communication, navigation and surveillance (CNS) satellites, including mobile uses and aeronautical uses, and of remote sensing, including meteorological satellites. Yet, the astronautical industry cannot be called mature at this stage. It is still a young, fledgling industry, subject to high insurance rates and to the real risk of launch failures, and dependent on direct and indirect subsidies. The commercial uses of outer space occur almost invariably in Earth orbit, and they also include the future commercial uses of the *International Space Station* (*ISS*), of which the major utilisation nevertheless remains scientific, technical and exploratory. Virtually all outer space activities beyond Earth orbit seem to be of an exploratory nature, although such activities may in the end have commercial spin-offs. Thus, outer space activities are commercialising, privatising, maturing, but they have not yet reached the maturity of ordinary or almost ordinary industries, such as the aeronautical industry, with which the astronautical industry has a number of things in common and with which, on occasions, it can very well be compared.[6] Several such comparisons will be made in this paper.

This text will examine how, gradually, the law prepares outer space for a competitive environment. Particular attention will be paid to international rules; to United States (U.S.) and

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[4] *Id* at art. I.

[5] *See* NICOLAS M. MATTE, AEROSPACE LAW 308-310 (Sweet and Maxwell / Carswell, 1969).

[6] *See* HAANAPPEL, *supra* note 1, at 12-13.

European Union (E.U.) legislation. The application of antitrust and competition laws will play an important role therein. So will the law pertaining to the allocation of scarce resources in outer space, where certain analogies are drawn from air law. The paper certainly does not claim to be exhaustive. It rather wants to outline a number of tendencies or possible tendencies.

II. THE EXTENSION OF THE COMPETENCE OF THE E.U. TO OUTER SPACE ACTIVITIES

Probably only the E.U. can boast to have outer space activities written into its basic constitutional document, or almost. Although the draft treaty establishing a Constitution for Europe was voted down in referenda in France and in The Netherlands in the year 2005, it will in all likelihood resurface in future versions, especially its less controversial provisions. It does not seem controversial to write that the "discovery of space" will be amongst the Union's objectives[7] and that the Union, in co-operation with member States, will carry out action and implement programmes "in the areas of research, technological development and space".[8]

Obviously, these new constitutional provisions look at exploration, at research and development, rather than at competition in the astronautical industry. Yet, they do not stand in the way of the application of the E.U. general competition rules to commercial, astronautical activities, as has already been advocated for, amongst others, the European satellite navigation project *Galileo*[9] –a joint undertaking of the E.U. and the Euro-

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[7] *See* Stephan Hobe, *Draft Text of the Articles of the Treaty Establishing a Constitution for Europe (Art. 3 and Art. 12)*, *in* LEGAL ASPECTS OF THE FUTURE INSTITUTIONAL RELATIONSHIP BETWEEN THE EUROPEAN UNION AND THE EUROPEAN SPACE AGENCY (Institute of Air and Space Law, University of Cologne / DLR, May 2003).

[8] *Id.*

[9] *See* F.G. von der Dunk, *Of Co-Operation and Competition: GALILEO as a Subject of European Law*, *in* LEGAL ASPECTS OF THE FUTURE INSTITUTIONAL RELATIONSHIP BETWEEN THE EUROPEAN UNION AND THE EUROPEAN SPACE AGENCY 47, 58-62 (Institute of Air and Space Law, University of Cologne / DLR, May 2003).

pean Space Agency (ESA) –and as the following will show in more detail.[10]

III. E.U. SATELLITE COMMUNICATIONS LAW

One area where there have been tremendously significant commercialisation developments, is E.U. satellite communications law. Communications law is almost a legal science in and of itself, of which satellite communications law is only one part. As so often with outer space related activities, the space segment operates in an environment of absence of national sovereignty, whereas most uses of this space segment are Earth oriented, thus operating in an environment of national sovereignties, some of which may be in conflict with each other.

In the E.U. telecommunications field, there have been many legislative initiatives, and this legislative road map should probably be seen in the light of a typical E.U. policy, namely the policy of creating a level "playing field" between de-monopolised and cross-border service providers.

Some highlights from amongst E.U. space communications laws are the following: Commission Directive 94/46 of 13 October 1994, sometimes called the "satellite Directive", abolishes special and exclusive rights for the provision of satellite services and satellite Earth station equipment within the E.U. In order to facilitate access to the space segment, then still largely in the hands of international satellite organisations, like Intelsat and Inmarsat, the Directive already stressed the need to comply by E.U. competition law in this area. This obligation was considerably strengthened by Article 7 of Commission Directive 2002/77 of 16 September 2002 on competition in the markets for electronic communications networks and services. Therefore, these directives opened up the market.

E.U. Directive 97/13 of 10 April 1997 contains a general licensing regime for telecommunication services and it covers the

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[10] *See also* F.G. von der Dunk, *Private Enterprise and Public Interest in the European "Spacescape", Towards Harmonized National Space Legislation for Private Space Activities in Europe*, 249-253 (unpublished doctoral dissertation, McGill University International Institute of Air and Space Law, Montreal, Canada, 1998).

licensing of satellite systems. Licensing may be necessary in more than one E.U. Member State. Although I have not encountered any concrete conflicts in the literature, it is to be recalled that under Article VI of the Outer Space Treaty, it is the "appropriate State Party" that is responsible for the authorisation and continuing supervision of private enterprise in outer space.

It is another matter whether such conflict can be avoided in the application of Council Regulation 1321/2004 of 12 July 2004 on the establishment of structures for the management of the European satellite radio navigation programmes,[11] *Galileo* and EGNOS. As mentioned before, *Galileo* is a programme of the E.U. and the European Space Agency (ESA), and it is intended to have significant private sector participation.[12] The augmentation system EGNOS involves the E.U., ESA and Eurocontrol. The Regulation sets up a Community agency, called the European GNSS Supervisory Authority, and the Authority will, amongst other things, have licensing duties vis-à-vis the private concession holder of *Galileo* and the economic operator of EGNOS.[13] Well, as already twice mentioned Article VI of the Outer Space Treaty of 1967 is concerned, this Treaty only allows adherence by States, but not by international organisations, like the E.U., although the Treaty does contain provisions, in Article XIII, on member States' duties with respect to their participation in international intergovernmental organisations.[14]

Are we seeing something here in space telecommunications law that we have been familiar with in air law for the past twenty years? Does the difficulty of the E.U. fitting into the worldwide Outer Space Treaty resemble the sometimes-encountered difficulty of E.U. air transport law fitting into the worldwide system of the Chicago Convention on International

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[11] Council Regulation (EC) 1321/2004 of 12 July 2004 on the establishment of structures for the management of the European satellite radio-navigation programmes, O.J. L 246, 20/07/2004, at 1-9.

[12] *See* Council Regulation 876/2002/EC on setting up the Galileo Joint Undertaking, 2002 O.J. (L 138/1).

[13] *Id.* at arts. 1, 2.

[14] On the European GNSS Supervisory Authority in general, *see* F.G. von der Dunk, *Towards Monitoring Galileo: The European GNSS Supervisory Authority in statu nascendi*, 55 ZLW 100 (2006).

Civil Aviation of 1944?[15] Of course, it also needs to be mentioned that as a matter, at least of policy, but probably also of law, the provisions on the new E.U. agency cannot be implemented until the above discussed extension of the competence of the E.U. to outer space activities has been effectuated.

IV. THE APPLICATION OF ANTITRUST LAW

Next is the application of antitrust law, by which I mean the competition laws of the United States. In the areas that are examined in this paper, it is especially the U.S. "essential facilities doctrine", which is interesting.[16] The origins of the doctrine seem to lie in the common law, where the common carrier has a duty to serve all who apply at reasonable rates. In legislation, it finds its origin mostly in Section 2 of the *Sherman Act*[17] that forbids monopolisation. The doctrine is used by the courts as well as by the regulators, such as the Federal Communications Commission and the Federal Trade Commission. It has a broad application, amongst others in transport and communications, and in the latter field especially in the areas of interconnectivity and interoperability. Although good definitions are difficult and perhaps even dangerous to give, the concept of "essential facility" is a monopolistic facility, non-discriminatory access to which must be given to all who apply and that at reasonable prices.

Under E.U. competition law, the essential facilities doctrine has been received by the EC Commission, perhaps not so much as an independent doctrine, but rather as an application of the abuse of a dominant position under Article 82 of the E.U. Treaty. Two early cases are actually aviation cases: in *London European* v. *Sabena*, Sabena was held to have abused its dominant position in refusing to give access to its computerised res-

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[15] Convention on International Civil Aviation, Dec. 7, 1944, art. 1, 61 Stat 1180, 1180 [hereinafter Chicago Convention].

[16] See Antonio Bavasso, *Essential Facilities: The Rise of an "Epithet" and the Consolidation of a Doctrine*, *in* COMMUNICATIONS IN EU ANTITRUST LAW: MARKET POWER AND PUBLIC INTEREST 221 (Kluwer Law International, 2003).

[17] Sherman Act, 26 Stat. 209 (1890).

ervation system to a competitor, London European;[18] and in *British Midland* v. *Aer Lingus*, the termination by Aer Lingus of its interline agreement with British Midland was equally held to amount to an abuse of a dominant position.[19] It is especially the latter case, the interlining case, which is interesting in the context of this paper, since interlining is air transport's version of interconnectivity in telecommunications.

In E.U. telecommunications law, it should be mentioned that the legislators have intervened in the area that we are discussing, for instance by the adoption of Directive 97/33 of the European Parliament and the Council of 30 June 1997 on interconnection in telecommunications with regard to ensuring universal service and interoperability through application of the principles of Open Network Provision (ONP).

V. THE APPLICATION OF COMPETITION LAW

Back to the application of E.U. competition law to satellite communications, we see, as early as the year 1991, the Commission published Guidelines on the Application of EEC Competition Rules in the Telecommunications Sector,[20] but, at that time, they did not mean much specifically in the filed of satellites, because that area would not be liberalised until 1994, in the so-called "satellite Directive", mentioned earlier. Since that time, however, the E.U. competition laws have also been applied to space telecommunications, provided that, according to the rules of the *Wood Pulp* decision of the European Court of Justice,[21] they can be "localised" as to implementation and/or effect in the E.U.

Even though, as already indicated, the Treaty on the E.U. does not yet refer to outer space activities, its Articles 81, 82 and 86 apply to the competitive relations of undertakings and

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[18] London European Airways PLC v. Sabena, Belgian World Airlines, Case IV/32.318, 1988 O.J. (L 317) 47.

[19] Air Lingus, Case IV/33.544, 1992 O.J. (L 96/34).

[20] Guidelines on the Application of EEC Competition Rules in the Telecommunications Sector, 91/C233/02, 1991 O.J. (C 233).

[21] A. Alstrom Osakeyhtio v Commission, 1988 E.C.R. 5193 (hereinafter *Wood Pulp* Case).

states, involved in such activities. In the space telecommunications area, these Articles, former Regulation 17,[22] and former Regulation 4064/89[23] have been applied to infringements, service arrangements, acquisitions, strategic alliances and joint ventures.[24] It is interesting to note by way of comparison that the space telecommunications sector was governed by the general Regulation 17, whereas the air transport sector had more protective aviation specific legislation apply to it, namely Regulations 3975 and 3976, as amended.[25] In that respect, space telecommunications were ahead of air transport in their relation with competition law. Today, of course, both are governed by the new Regulation 1/2003 of 16 December 2002 on the implementation of the rules on competition laid down in Articles 81 and 82 of the Treaty.

What both industries also have in common, is that they have a lot of liberalisation and user oriented legislation apply to them, in the space sector notably ONP rules, with the consequence that both industries are heavily regulated, *ex ante*, transport and telecommunications law, and *ex post facto*, competition law. This is a sort of double legislative jeopardy, which seems unfair to a liberalised industry. In mature industries, it seems, *ex ante* regulation should be limited to licensing, with the remainder of the regulation, especially competition law, applying principally on an *ex post facto* basis. Licensing today may also include elements, it seems, of industry *self* regulation.

VI. CONFLICTS OF LAW

The major players in outer space and its applications are the U.S., the E.U., the Russian Federation, Japan, and in the future no doubt China. And, of course, there is a growing number of others, such as Argentina, Australia, Brazil, Canada –of long standing, and with associate membership in ESA– India and South Africa. They all have their own legal and regulatory

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[22] The general competition law implementing Regulation, now superseded.

[23] The Merger Regulation, Commission Regulation 4064/89 (superseded).

[24] See Colin D. Long, TELECOMMUNICATIONS LAW AND PRACTICE 285 (Sweet & Maxwell, 2nd ed., 1995).

[25] For these Regulations, *see* HAANAPPEL, *supra* note 1, at 138-140.

systems, between which conflicts cannot always be avoided, given also the no-frontier character of outer space activities. Conflicts are there to be solved. They can be solved bilaterally or multilaterally, judicially or extra-judicially, in private or in public law.

The opinion is often heard that participants in outer space activities should have their own space-specific dispute settlement system.[26] I am not so sure, especially once the astronautical industry has become mature and can, like other industries, avail itself of generally available dispute settlement systems. I will revert to this question later, under the heading of the World Trade Organisation (WTO).

Three areas of potential conflict need special mention: mergers, subsidies, and the *ISS*. The General Electric / Honeywell Merger case show to what extent antitrust and competition law policies in the U.S. and in the E.U. may clash.[27] The U.S. approves a concentration; the E.U. disapproves it. Tomorrow the reverse may be true. As perhaps in the future more astronautical industry concentration takes place, will such policy disputes also affect outer space endeavours?

Both the U.S. and the E.U. will continue, at least for the foreseeable future, to subsidise outer space activities. Both legally can,[28] but disputes may arise. The Arianespace monopoly may be a good example,[29] and one wonders whether the Airbus-Boeing subsidy dispute might serve as an appropriate precedent.[30]

Also, the question has been asked whether commercial activities involving the *ISS* would require a special dispute set-

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[26] Currently, such system only exists within the Liability Convention in the form of a Claims Commission: *see* Convention on International Liability for Damage Caused by Space Objects, *in force* 1 September 1972, 24 U.S.T. 2389, TIAS 7762, 961 UNTS 187 [hereinafter Liability Convention].

[27] *See, The GE / Honeywell Merger Case: Reaching the Limits of International Competition Policymaking*, 2 (no. 12) GERMAN L.J. 1-10 (2001).

[28] In the E.U., only member States are under restrictions with respect to State aids, not the Union itself.

[29] *See* Nathanael A. Horsley, *The Arianespace Monopoly, EU Competition law, and the Structure of Future European Launch Markets*, 20 AIR & SPACE L. 87 (2005).

[30] *See* Ruwantissa Abeyratne, *The Airbus-Boeing Subsidies Dispute – Some Preliminary Legal Issues*, 30 AIR & SPACE L. 379 (2005).

tlement mechanism. One author has, it seems to me appropriately, suggested WTO for this purpose.[31]

Finally, now that it has become evident that there will be several CNS systems in the world, rather than just one under the umbrella of the International Civil Aviation Organization (ICAO), compatibility between these systems must be ensured. In that respect it is hopeful to note that the U.S. and the E.U. have signed an agreement with respect to the frequency allocation and interoperability of the US Global Positioning Satellite System (GPS) and Europe's *Galileo*.[32]

VII. THE ALLOCATION OF SCARCE RESOURCES IN OUTER SPACE: COMPARISON OF ITU / WARC AND ICAO / IATA RULES

A truism as it may be, outer space is infinitely larger than air space, and hence it is likely to have fewer scarce resources than air space. Yet, for over thirty years now, the equatorial geostationary orbit has been declared such a limited natural resource. It is the preferred location for communication satellites intended to have worldwide coverage. Through the International Telecommunication Union (ITU) and the Radio Regulations of its World Administrative Conferences (WARCs), frequencies in this orbit are allotted on a first-come, first-served basis, without, however, creating any right of permanency to these frequencies.[33] This can be compared with the best known system of allocation of scarce resources in air space, the slot allocation system of the Scheduling Guidelines and Conferences of the International Air Transport Association (IATA),[34] which find

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[31] *See* A.C. Swan, *Competition Policy and the International Space Station*, *in* 16 STUDIES IN AIR AND SPACE LAW, "PROJECT 2001" – LEGAL FRAMEWORK FOR THE COMMERCIAL USE OF OUTER SPACE, RECOMMENDATIONS AND CONCLUSIONS TO DEVELOP THE PRESENT STATE OF THE LAW 375 (K.–H. Böckstiegel, ed.,2002).

[32] *See* Agreement on the Promotion, Provision and Use of GALILEO and GPS Satellite-Based Navigation Systems and Related Applications, *available at* ec.europa.eu/dgs/energy_transport/galileo/documents/doc/2004_06_21_summit_2004_en.pdf (last visited Aug. 15, 2006).

[33] *See* HAANAPPEL, *supra* note 1, at 24, 39-40. This system of allocation is not to be confused with questions of space traffic management, safety questions that, *inter alia*, seek to avoid space debris from doing damage to space objects.

[34] HAANAPPEL, *supra* note 1, at 153-156. *See also*, Regulation (EC) No 793/2004 of the European Parliament and of the Council of 21 April 2004 amending Council Regula-

their legal basis in the uniformity requirement of Article 15 the Chicago Convention.[35]

Whereas first-come, first-served is not enunciated in so many words in the IATA rules, it amounts to that, as long as there are sufficient arrival and departure slots at airports and airport runways available. Where the biggest difference between ITU and IATA rules lie, are in the "absence of permanency" of ITU versus the grandfather or historical precedence rights of the IATA system. It is submitted that the IATA system is, from the point of view of law and economics, superior to the ITU system in that it recognises that financial investments in slots have been made that need some permanency so as to keep their economic value. Granted that the original ITU rules of the 1970s were made under pressure from certain equatorial nations claiming sovereignty rights in the geostationary orbit, this does not mean that they should stay the same forever.[36] The need of "have not" states for frequencies can also be met by the setting up of pools of unused or newly created slots, like in the IATA system. In order to avoid overbooking of frequency allocations, use-it-or-loose it rules can be adopted. From the U.S. domestic system of airport slot allocation, the occasional use of slot auctions or lotteries could be copied.

VIII. THE FUTURE ROLE OF THE WTO: IS GLOBALISATION THE ANSWER?

Getting closer to the end of this paper, and getting closer to the main theme of the 20th Anniversary Conference of Leiden's International Institute of Air and Space Law: is globalisation the answer, especially in a WTO context? Compared with air transport, where the General Agreement on Trade in Services (GATS) only has a rather insignificant Annex on Air Transport Services apply to it, communications, including space communications, are much more extensively covered by the umbrella of

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tion (EEC) No 95/93 on common rules for the allocation of slots at Community airports, *available at* http://eur-lex.europa.eu/LexUriServ/site/en/oj/2004/l_138/l_13820040430en00500060.pdf (last visited Aug. 15, 2006).

[35] Chicago Convention, *supra* note 15.

[36] See references in HAANAPPEL, *supra* note 1, at 24, 39-40.

the WTO. Together, the Annex on Telecommunications to the GATS Agreement of April 1994 and the Fourth Protocol to the GATS Agreement of April 1996 / February 1997, dealing with basic telecommunications, provide that participating States commit themselves to allow foreign satellite communication operators to offer their services on a reciprocal, non-discriminatory basis in their countries. In addition, there is the potential application of the TRIPS agreement, dealing with intellectual property rights, and of the WTO dispute settlement rules to space communications. Future application of a "Standardisation Code"[37] and of the Agreement on Subsidies and Countervailing Duties (SCM) is also not excluded.

Indeed, the globalisation of WTO seems the way forward in space law, not only for telecommunications, but also for other privatised commercial outer space activities. In this respect, commercial outer space activities have an edge on air transport services, where bilateral air transport and services agreements are so enshrined in sixty years of tradition that they are hard to get rid off in favour of a WTO regime, and that certainly as long as the United States does not seem willing to abandon the system of bilateral agreements in international aviation.

Finally, a note to say that a GATS Party making a specific commitment in the field of outer space activities seems to accord very well with the "appropriate State Party" in Article VI of the Outer Space Treaty.

IX. CONCLUDING REMARKS

It seems that the astronautical industry is rapidly becoming a mature, privatised or at least commercialised industry and that the law is adjusting to that situation very well. However, like in aeronautics, in astronautics governments will never leave the industry really alone, and this is because of national defence and national security considerations. After all, one and

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[37] Code of Good Practice for the Preparation, Adoption and Application of Standards, *in* Agreement on Technical Barriers to Trade, Dec. 15, 1993, *available at* http://www.iso.org/iso/en/comms-markets/wto/pdf/tbt-a3.pdf#search=%22Code%20of%20Good%20Practice%20for%20the%20Preparation%2C%20Adoption%20and%20Application%20of%20Standards%22 (last visited Aug. 15, 2006).

the same aircraft, or one and the same satellite or satellite system can fulfill both civil and military purposes. For aircraft, the Civil Reserve Air Fleet (CRAF) programme in the U.S. would be a good example. For spacecraft, let's, for instance, not forget that GPS is in principle a military programme of the American Department of Defence (DOD) with civil applications, or that remote sensing satellites can be used both for military and civil purposes.[38]

[38] In which case the U.N. Principles on Remote Sensing only apply to some of the civil, but not to the military uses of remote sensing satellites: *see* HAANAPPEL, *supra* note 1, at 159-160.